SAPTHAGIRI COLLEGE OF ENGINEERING, BENGALURU-560 057.
(Affiliated to VTU, Belagavi & Approved by AICTE, New Delhi)
DEPARTMENT OF MATHEMATICS
QUESTION BANK (I SEM)
ACADEMIC YEAR: 2022-23 (ODD SEMESTER)

MODULE-5: Linear Algebra

Sub: MATHEMATICS-I **Sub Code: BMATE/S/C/M101**

1. Find the rank of the Matrix
$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ by applying elementary Row transformations.

2. Find the rank of the Matrix **A**$=\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$
by applying elementary Row transformations.

3. Find the rank of the Matrix
$A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$ by applying elementary Row transformations.

4. Find the rank of the Matrix
$A = \begin{bmatrix} 1 & 0 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ by applying elementary Row transformations.

5. Test for consistency and solve:
i) $5x_1 + x_2 + 3x_3 = 20;\ 2x_1 + 5x_2 + 2x_3 = 18; 3x_1 + 2x_2 + x_3 = 14$.
ii) $x + y + z = 6; x - y + 2z = 5;\ \ 3x + y + z = 8.$
iii)$x + 2y + 3z = 14;\ \ 4x + 5y + 7z = 35;\ \ 3x + 3y + 4z = 21.$

6. For what values of λ and μ the system of equations.
$x + y + z = 6; x + 2y + 3z = 10; x + 2y + \lambda z = \mu$. has
i) Unique solution ii) Infinite number of solutions iii) No solution.

7. Solve the system of equations by Gauss-Elimination Method
i) $x + y + z = 9;\ x - 2y + 3z = 8; 2x + y - z = 3.$
ii) $2x + y + 4z = 12;\ 4x + 11y - z = 33;\ 8x - 3y + 2z = 20.$
iii) $x_1 - 2x_2 + 3x_3 = 2;\ 3x_1 - x_2 + 4x_3 = 4;\ 2x_1 + x_2 - 2x_3 = 5$.

8. Solve the system of equations by Gauss-Jordan Method:
i) $x + y + z = 9;\ x - 2y + 3z = 8;\ 2x + y - z = 3.$
ii)$2x + 5y + 7z = 52;\ 2x + y - z = 0;\ x + y + z = 9.$
iii) $2x_1 + x_2 + 3x_3 = 1;\ 4x_1 + 4x_2 + 7x_3 = 1;\ 2x_1 + 5x_2 + 9x_3 = 3$

9. Solve the system of equations by Gauss-Seidel Method:
i) $20x + y - 2z = 17;\ 3\,x + 20y - z = -18,\ 2x - 3y + 20z = 25.$
Carry out 3 iterations taking the initial approximation to the solutions as (1,0,0).

ii) $5x + 2y + z = 12;\ x + 4y + 2z = 15,\ x + 2y + 5z = 20.$
Carry out 4 iterations taking the initial approximation to the solutions as (1,0,3)

iii) $10x + y + z = 12;\ x + 10y + z = 12,\ x + y + 10z = 12.$

10. Using Rayleigh's power method find the largest Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ with $X^{(0)} = [1 \quad 1 \quad 1]^T$ as the initial Eigen vector. (Perform 7 iterations)

11. Using Rayleigh's power method find the largest Eigen value and the corresponding Eigen vector of the matrix A=$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$with $X^{(0)} = [1 \quad 1 \quad 1]^T$ as the initial Eigen vector. (Perform 6 iterations)

12. Using Rayleigh's power method find the largest Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ with $X^{(0)} = [1 \quad 0.8 \quad -0.8]^T$ as the initial eigen vector. (Perform 5 iterations)

13. Using Rayleigh's power method find the largest Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with $X^{(0)} = [1 \quad 1 \quad 1]^T$ as the initial eigen vector. (Perform 5 iterations)